Retail Fashion Sales Analysis

Project Focus: Sales Analysis
Project Kind: Data Analyst Project

Dataset Info:
Title: Messy Retail Fashion Data
Source: Kaggle
Link: https://www.kaggle.com/datasets/vanpatangan/retail-fashion-data
No. of CSV files: 4

Dataset Problem:
1. Messy Values
2. Invalid Keys
3. Inconsistent Formats

Business Question:
1. Customer Strategy & Marketing Optimization
    a. Which high-value customer segments are forecasted to have the highest churn probability over the next 90 days?
    b. What proactive marketing interventions should we launch today to alter the income?
    - CSV: customer_data.csv, sales_data.csv
2. Inventory & Markdown Management
    a. What is the forecasted demand for seasonal apparel categories by size and color over the next quarter?
    b. How should we structure our automated markdown timeline to liquidate slow-moving stock before its value plummets?
    - CSV: sales_data.csv, product_data.csv
3. Supply Chain & Supplier Performance
    a. Which specific variants are predicted to face stockouts or critical inventory deficits next month?
    b. How should we reallocate our upcoming procurement orders among top-performing vendors to prevent this?
    - CSV: product_data.csv, sales_data.csv
4. Omnichannel & Store Performance Benchmarking
    a. What is the total income per location?
    b. Which physical stores should we proactively plan to transition into localized omnichannel fulfillment centers?
    - CSV: store_data.csv, sales_data.csv

Data Metrics:
1. Customer Segment and Churn Rate
2. Demand Forecasts per Season
3. Stock Demand
4. Revenue Forecast for a year

Tech Stack: Power BI

Type of Analysis: Prescriptive Analysis

Analysis Steps:

1. ETL and EDA

CSV: customer_data			
Row Count (before cleaning): 25,000			
Row Count (after cleaning): 25,000			
Column	**Data Type**	**Problem**	**Transformation/ Cleaning**
customer_id	str	None. No duplicates and unique.	None
age	int	None. No missing values.	None
gender	str	Missing Values = “???” (4 items)	Changed “???” to “Other”
city	str	None	None
email	str	Some values are missing	Left it blank. Cannot delete missing values as it will delete the record completely.

CSV: product_data			
Row Count (before cleaning): 50,000			
Row Count (after cleaning): 50,000			
Column	**Data Type**	**Problem**	**Transformation/ Cleaning**
product_id	str	None. No duplicates.	None
category	str	Missing Values = “???” (9 items)	Changed “???” to “Other”. Cannot be deleted as it will delete the record completely.
color	str	Missing Values = 990 items	Replaced the missing value with “Other”
size	str	None. No missing values.	None
season	str	None. No missing values.	None
supplier	str	None. No missing values.	None
cost_price	float	None. No missing values.	None
list_price	float	None. No missing values.	None

CSV: sales_data			
Row Count (before cleaning): 50,000			
Row Count (after cleaning): 50,000			
Column	**Data Type**	**Problem**	**Transformation/ Cleaning**
transaction_id	str	None. No duplicates.	None
date	date	None. No missing values.	None

product_id	str	None. No missing values.	None
store_id	str	Invalid store_id: “S999” (Count = 200), Valid values from store_data: S001, S002, S003, S004, S005	Cannot delete invalid values as it will skew the results.
customer_id	str	Missing Values = 1844 items	Cannot delete missing values as it will skew the results.
quantity	str	None. No missing values.	None
discount	float	“null” Values = 2583 items	Replaced “null” values with zero
returned	int	None	None

CSV: store_data			
Row Count (before cleaning): 5			
Row Count (after cleaning): 5			
Column	**Data Type**	**Problem**	**Transformation/ Cleaning**
store_id	str	None	None
store_name	str	None	None
region	str	None	None
store_size_m2	int	None	None

2. Data Modeling

Fact Table	**Dimensions**	**Column**	**Cardinality**
sales_data	customer_data	customer_id	Many-to-One
	product_data	product_id	Many-to-One
	store_data	store_id	Many-to-One

3. Feature Engineering

Calculated Columns		
Tool: **DAX**		
Column	**Code**	**Factors Considered**
total_amount	`= (sales_data[quantity] * RELATED(product_data[list_price])) * (1 - sales_data[discount])`	

customer_segment	`= VAR customer_sales =`
	`CALCULATE(SUM(sales_data[total_amount]))`
	`RETURN`
	`    SWITCH(`
	`        TRUE(),`
	`        customer_sales >= 2500, "High-Value",`
	`        customer_sales >= 1000, "Mid-Value",`
	`        "Low-Value"`
	`    )`
		Highest Customer Total Amount = $3.2K

gross_revenue	`= sales_data[quantity] *` `RELATED(product_data[list_price])`	Calculate the gross before discounts
discount_loss	`= -1 * sales_data[gross_revenue] *` `sales_data[discount]`	Calculate the discount loss

discount_tier	`SWITCH(`
	`    TRUE(),`
	`    sales_data[discount] == 0, "Full Price",`
	`    sales_data[discount] == 0.10, "Low Discount",`
	`    sales_data[discount] == 0.20, "Mid Discount",`
	`    "Deep Clearance"`
	`)`
		For markdown tracking

Calculated Measures		
Tool: **DAX**		
Measure	**Code**	**Factors Considered**
sum_total_amount	`= SUM(sales_data[total_amount])`	KPI
total_customers	`= DISTINCTCOUNT(customer_data[customer_id])`	KPI
most_recent_dataset_date	`= CALCULATE(MAX(sales_data[date]), ALL(sales_data))`	Churn Metric
last_customer_purchase	`= MAX(sales_data[date])`	Churn Metric
days_since_last_purchase	`= DATEDIFF([last_customer_purchase],` `[most_recent_dataset_date], DAY)`	Churn Metric

customer_churned	`= SUMX(`
	`    customer_data,`
	`    IF([days_since_last_purchase] >= 90, 1, 0)`
	`)`
		Churn Metric

churn_rate_%	`= DIVIDE(`
	`    [customers_churned],`
	`    [total_customers],`
	`    0`
		Churn Metric

	)	
churned	`= IF([days_since_last_purchase] >= 90, "Yes", "No")`	Churn Filter
total_income	`= SUM(sales_data[total_amount])`	Markdown Metric
sum_gross_revenue	`= SUM(sales_data[gross_revenue])`	Markdown Metric
sum_discount_loss	`= SUM(sales_data[discount_loss])`	Markdown Metric

forecasted_demand	`= VAR total_quantity = SUM(sales_data[quantity])`
	`VAR months_in_dataset =`
	`DISTINCTCOUNT(sales_data[date].[Month])`
	`RETURN`
	`    DIVIDE(total_quantity, months_in_dataset, 0)`
		Demand Metric

net_inventory_deficit	`= VAR total_units = SUM(sales_data[quantity])`
	`VAR total_months =`
	`DISTINCTCOUNT(sales_data[date].[Month])`
	`VAR monthly_demand_rate = DIVIDE(total_units,`
	`total_months, 0)`
	`VAR target_baseline_stock = monthly_demand_rate * 1.5`
	`RETURN`
	`    target_baseline_stock - monthly_demand_rate`
		Demand Metric

variants_at_risk_count	`= CALCULATE(`
	`    DISTINCTCOUNT(product_data[product_id]),`
	`    FILTER(`
	`        product_data,`
	`        [net_inventory_deficit] < 0`
	`    )`
	`)`
		Demand Metric

4. Data Visualization

Page Name: **customer&strategy**			
Chart Name	**Chart Type**	**Fields**	**Reason**
Predicted Churn Rate by Customer Segment	Clustered Bar Chart	Y-axis: customer_segment; X-axis: churn_rate percentage	Easily view the churn rate of the different customer segments
Total Spend by Predicted Churn Rate % per Customer Segment	Scatter Chart	Values: customer_segment; X-axis: total_spend; Y-axis: churn_rate_%; Legend: customer_segment; Size: total_income	Easily view the correlation between the total spend and churn rate percentage of the different customer segments and their total customers as the size of the points
Customer Details	Table	Columns: customer_id, customer_segment,	Easily view which customers and their total spend are

		total_spend	possibly churning
Predicted Churn Rate %	Card	Value: churn_rate_%	Easily view the churn rate
Total Customers	Card	Value: total_customers	Easily view the total customers
Predicted Total Customers Churned	Card	Value: customers_churned	Easily view the customers churned
Customer Segment	Slicer	Value: customer_segment	Easily filter by customer segment
Churn Possibility	Slicer	Value: churn_possibility	Easily filter by churn possibility
Year	Slicer	Field: date (Year)	Easily filter by year

Page Name: **inventory&markdown**			
Chart Name	**Chart Type**	**Fields**	**Reason**
Average of Quantity by Year and Quarter	Line Chart	X-axis: date (Year and Quarter); Y-axis: Average of Quantity	Easily visualize the quantity by Product Category and Size
Total Gross Revenue and Total Discount Loss by Discount Tier	Waterfall Chart	Category: discount_tier; Y-axis: sum_gross_revenue; Tooltips: sum_discount_loss	Easily visualize the Gross Revenue and Discount Loss per Discount Tier
Total Net Income	Card	Value: Sum of total_income	Easily view Total Income
Total Customers	Card	Value: total_customers	Easily view Total Customers
Total Gross Revenue	Card	Value: sum_gross_revenue	Easily view Gross Revenue
Total Discount Loss	Card	Value: sum_discount_loss	Easily view Discount Loss
Size	Slicer	Value: size	Easily filter by Size
Product Category	Slicer	Value: category	Easily filter by Category
Year	Slicer	Field: date (Year)	Easily filter by Year

Page Name: **supply&supplier**			
Chart Name	**Chart Type**	**Fields**	**Reason**
Forecasted Demand and Net Inventory Deficit by Category, Color, Size	Matrix	Rows: category, product_id; Columns: color, size; Values: forecasted_demand, net_inventory_deficit	Easily view the Forecasted Demand and Net Inventory Deficit by different variables
Forecasted Demand and Net Inventory Deficit by Product ID	Stacked Bar Chart	Y-axis: product_id; X-axis: forecasted_demand, net_inventory_deficit	Easily view which Product ID has the highest Forecasted Demand and Net Inventory Deficit
Analysis of Total	Decompositi	Analyze: total_income;	Easily analyze which Supplier

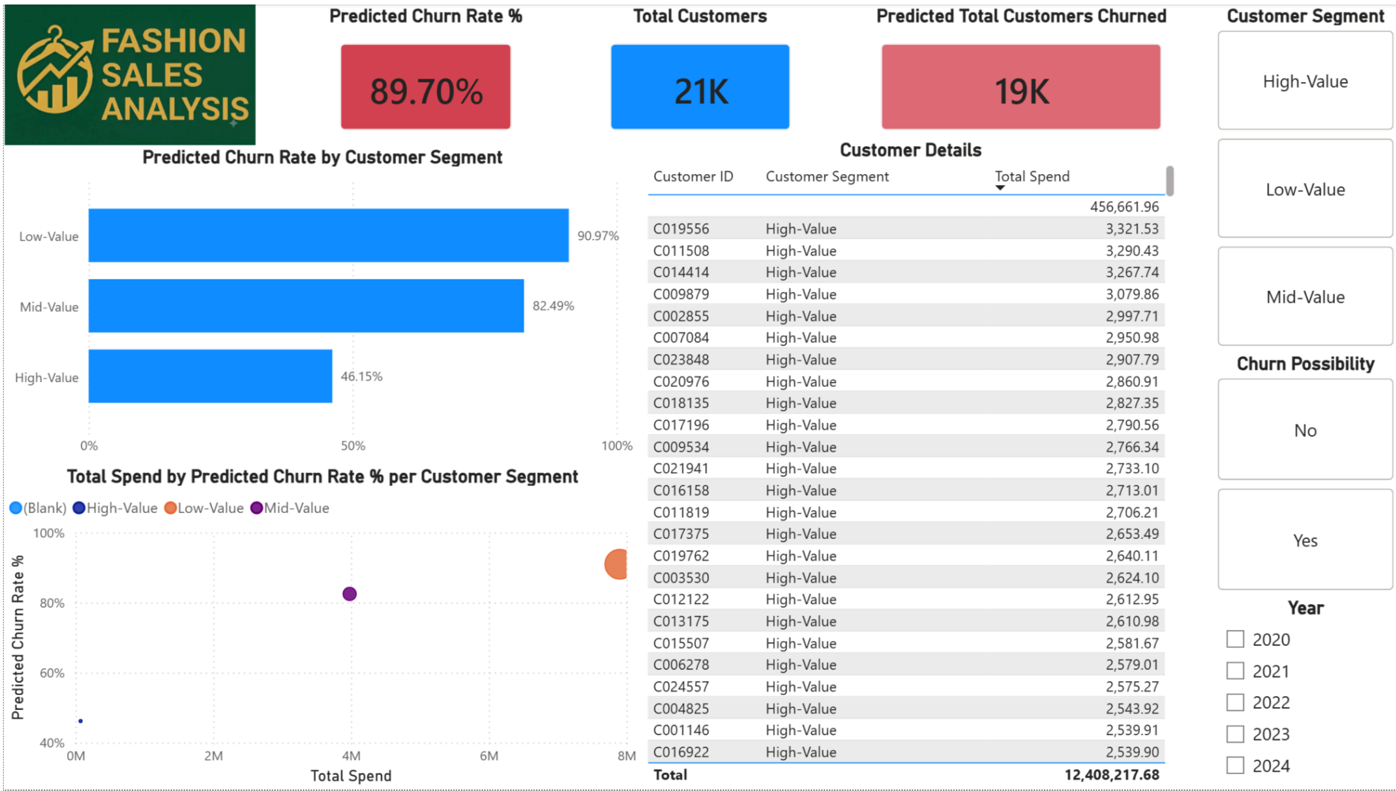
Income by Supplier, Category, & Product ID	on Tree	Explain by: supplier -> category -> product_id	has the highest to lowest income and expand by Category and Product ID
Total Net Income	Card	Value: Sum of total_income	Easily view Total Income
Forecasted Demand	Card	Value:forecasted_demand	Easily view Forecasted Demand Next Month
Net Inventory Deficit	Card	Value: net_inventory_deficit	Easily view safety buffer gap between a target safety stock baseline and actual monthly sales velocity
Low Inventory Products	Card	Value: variants_at_risk_count	Easily view how many products have low inventory stocks
Size	Slicer	Value: size	Easily filter by Size
Color	Slicer	Value: color	Easily filter by Color
Product Category	Slicer	Value: category	Easily filter by Category
Year	Slicer	Field: date (Year)	Easily filter by Year

Page Name: **omnichannel&store**			
Chart Name	**Chart Type**	**Fields**	**Reason**
Total Income by Store ID	Line chart	X-axis: date; Y-axis: total_income; Legend: store_id	Easily view the Total Income of the different Stores
Total Income by Store Size	Scatter chart	X-axis: store_size_m2; Y-axis: total_income; Legend: store_name; Size: total_income	Easily view the correlation between Store Size and Total Income of the different stores
Total Income by Region	Bar chart	Y-axis: region; X-axis: total_income	Easily compare the Total Income by Region
Total Net Income	Card	Value: Sum of total_income	Easily view Total Income
Store ID	Slicer	Value: store_id	Easily filter by Store ID
Region	Slicer	Value: region	Easily filter by Region
Date	Slicer	Value: date	Easily filter by Date

Answers:

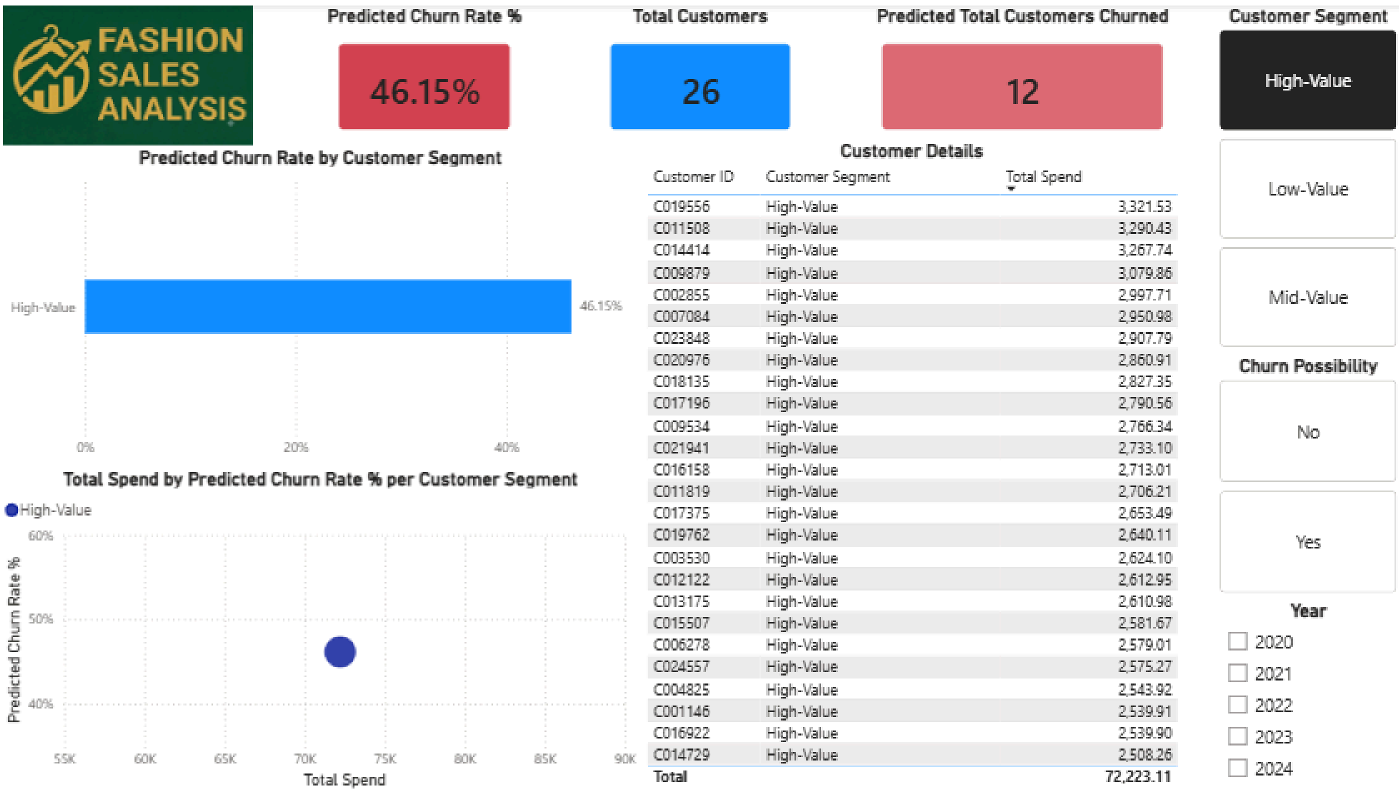
1. Customer Strategy & Marketing Optimization
    a. Which high-value customer segments are forecasted to have the highest churn probability over the next 90 days?

General Overview:

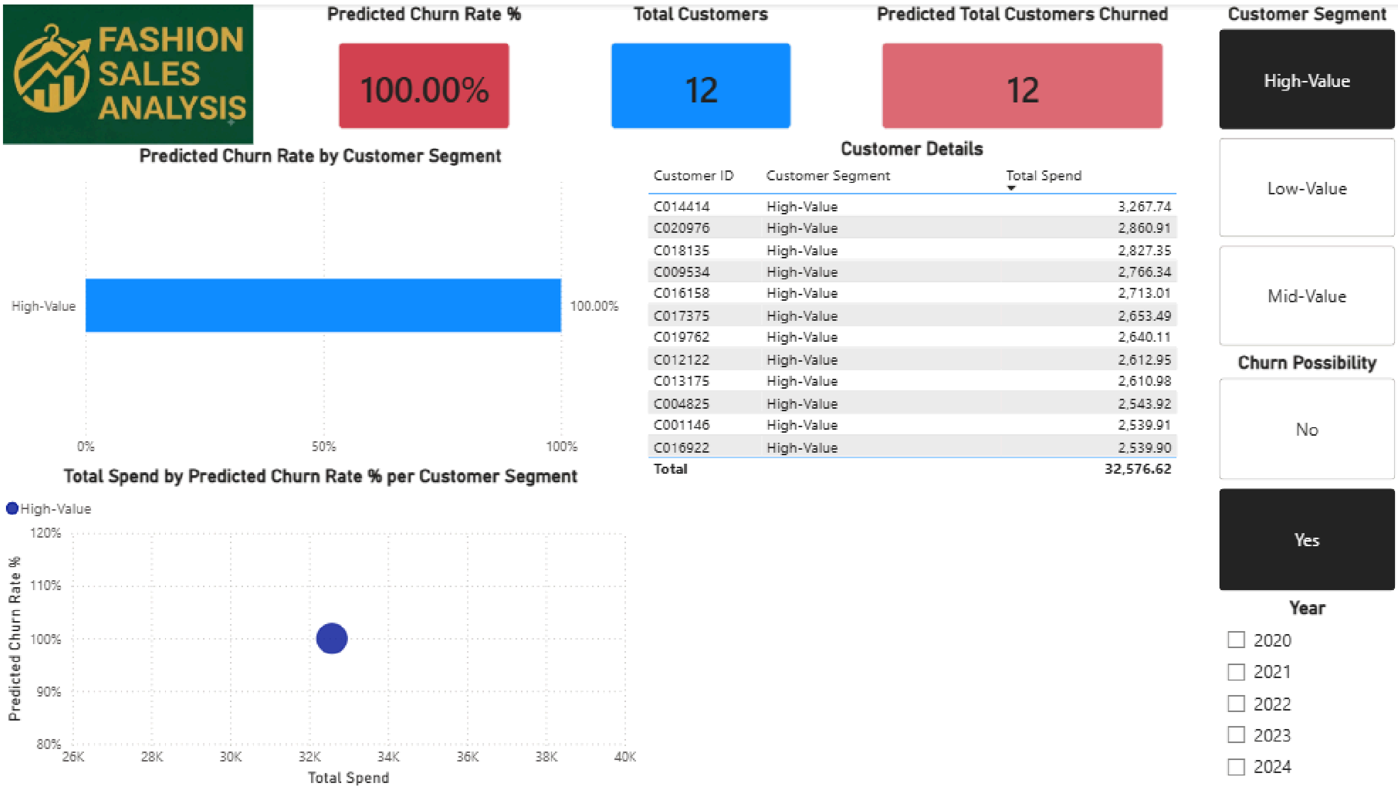


- Overall, there is a predicted churn rate of almost 90%. The churn rate is calculated by identifying which customers have not purchased since their last purchase 90 days ago.
- As can be seen in the visual, almost 19,000 customers are predicted to churn.
- Low-Value Customers have the highest churn rate while High-Value Customers have the lowest churn rate. However, even if the Low-Value Customers have the highest churn rate, they are also the highest spenders.

Analyzing High-Value Customers:



- High-Value Customers have a churn rate of 46.15%.
- 12 out of 26 high value customers are predicted to churn.



- The third highest spender among the high-value customer segment, Customer ID: C014414 is predicted to churn. The customer is followed by other customers with spending less than 2k but not lower than 2.5k.

b. What proactive marketing interventions should we launch today to alter the income?

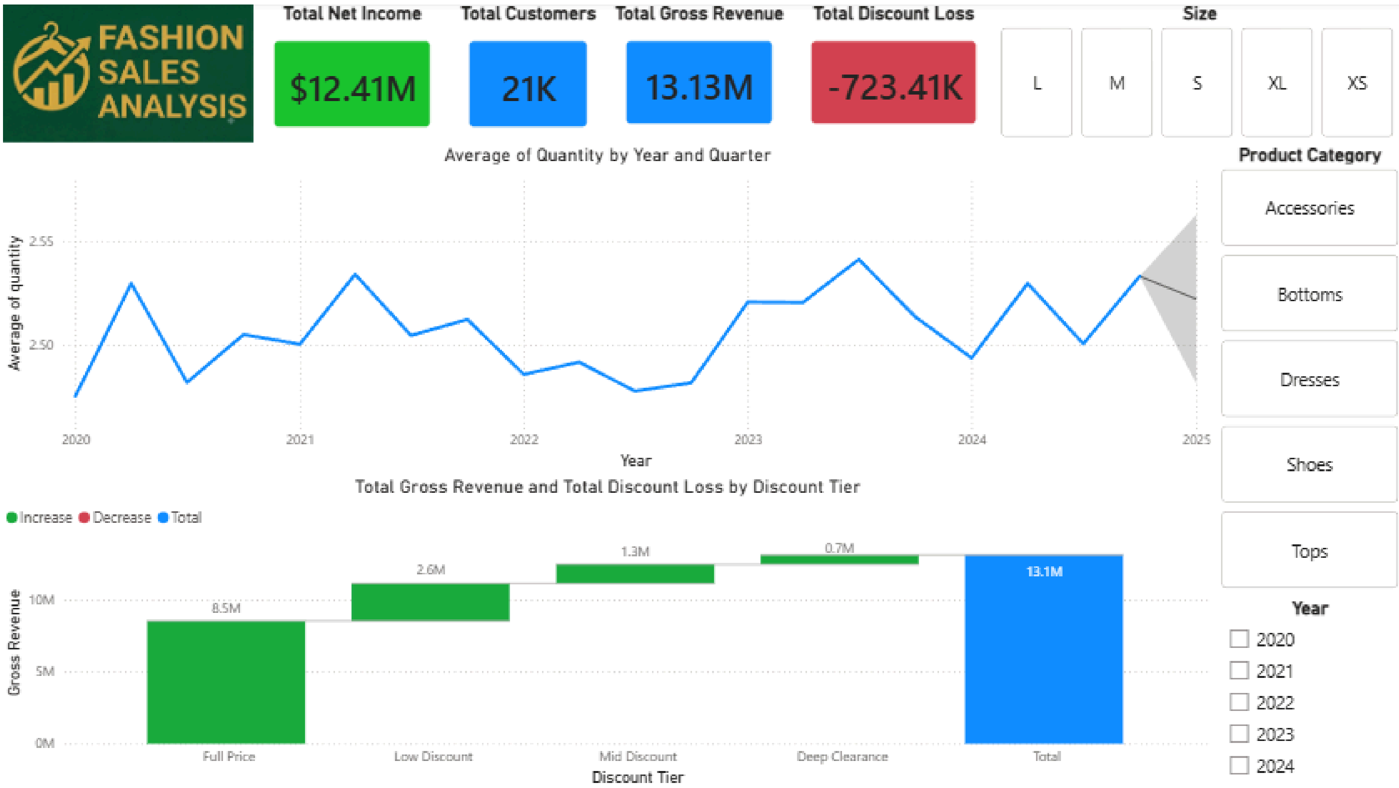
Possible interventions:

- Personalized Outreach: Immediately launch a 1-on-1 direct outreach campaign (e.g. email or dedicated clienteling) offering perks (e.g. early access ot new seasonal collection) or an exclusive VIP discount to the 12 flagged high-value customers.
- Spend-Threshold Reactivation: Offer custom, tiered spend-an-save incentives (e.g. $50% off your next purchase over $250) specially designed to re-engage customers before they hit the 90-day inactive mark.
- Feedback & Win-Back Surveys: Send automated “We miss you” surveys combined with a small store credit before they hit the 90-day inactive mark.
- 60-Day Warning Automation: Implement automated email workflows triggered at Day 60 of inactivity to engage customers before reaching the 90-day churn threshold.

2. Inventory & Markdown Management

a. What is the forecasted demand for seasonal apparel categories by size and color over the next quarter?

General Overview:



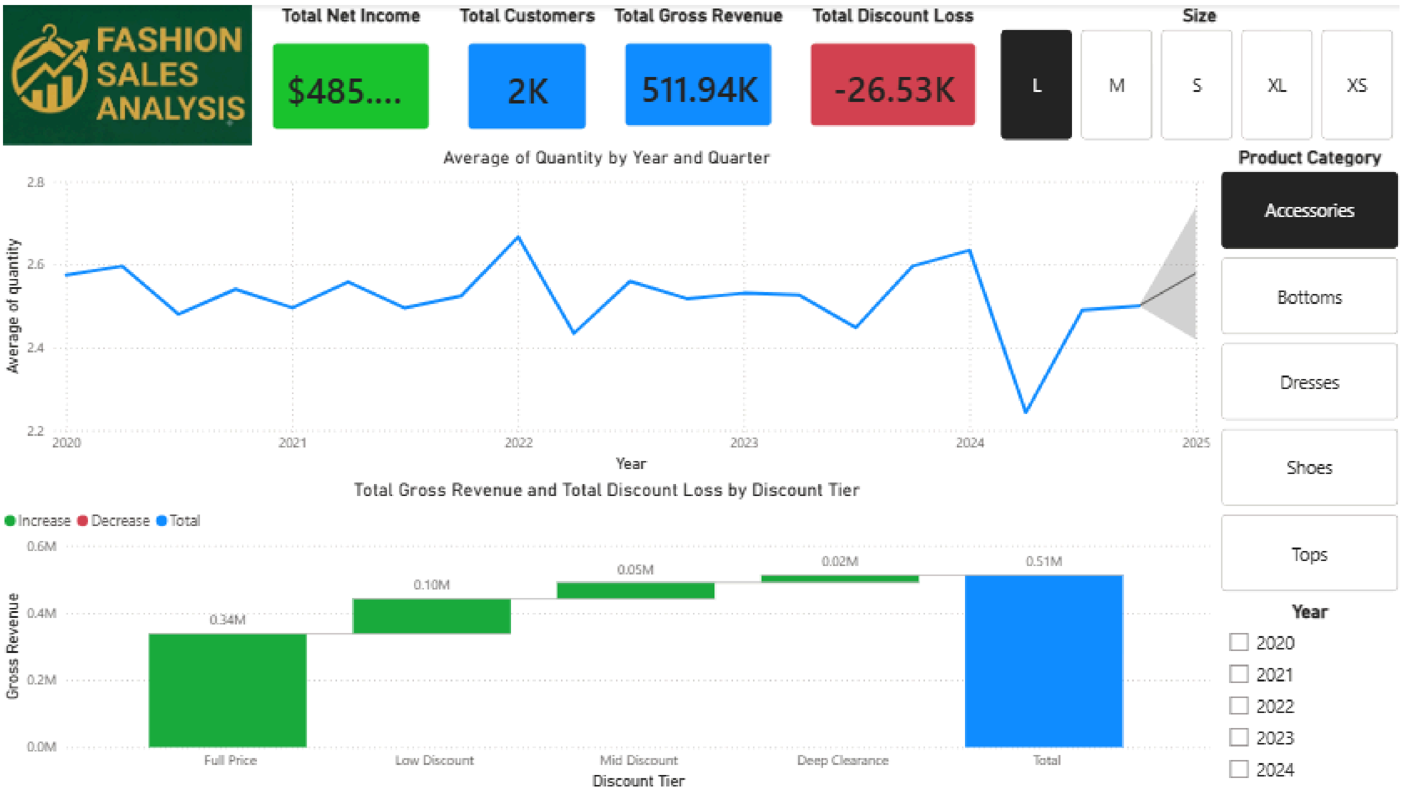
- The predicted average quantity of products sold for the first quarter of 2025 is expected to be lower.

Product Category and Sizes Analysis:

- Only included sizes that are forecasted to be in-demand next quarter.

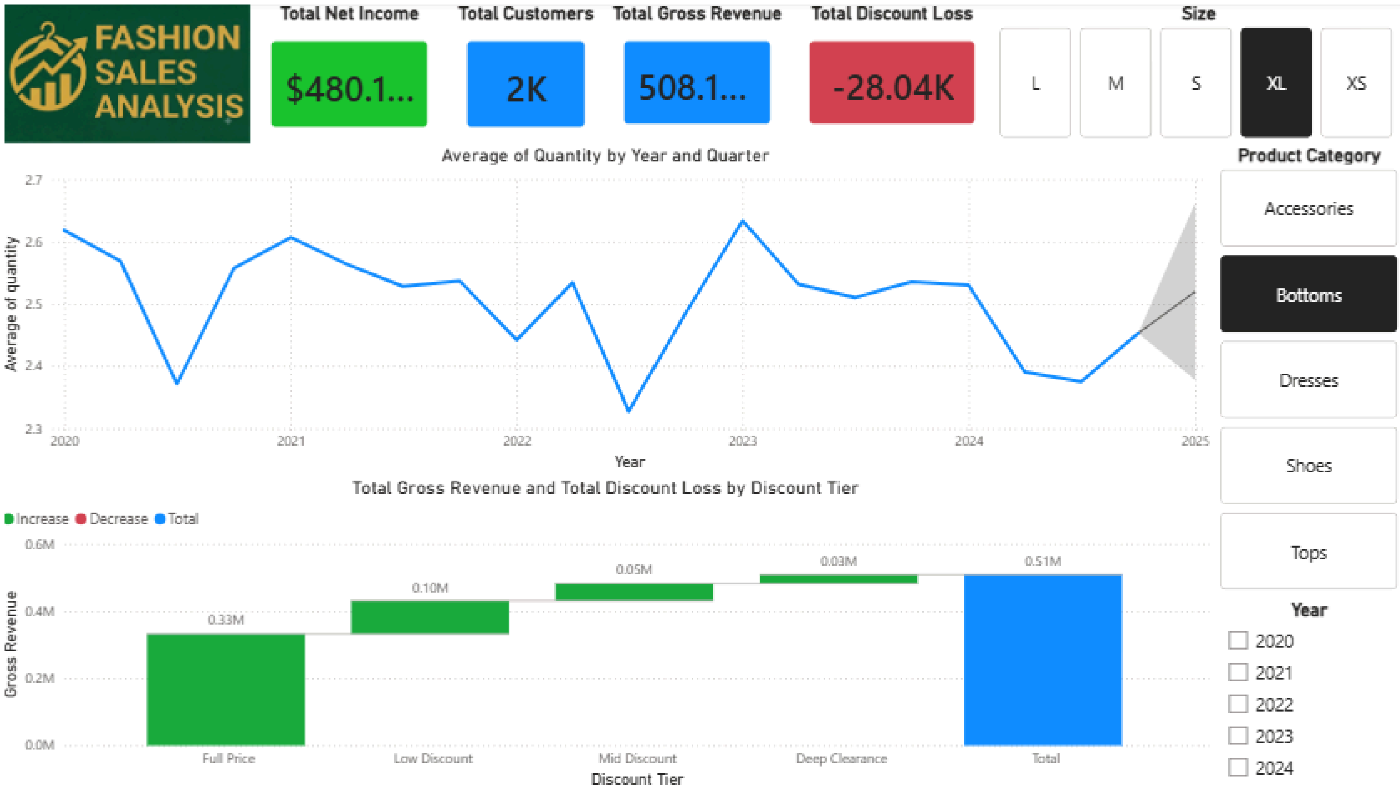
Accessories

- Stock up the following sizes: L, S, XL



Bottoms

- Stock up the following sizes: XL

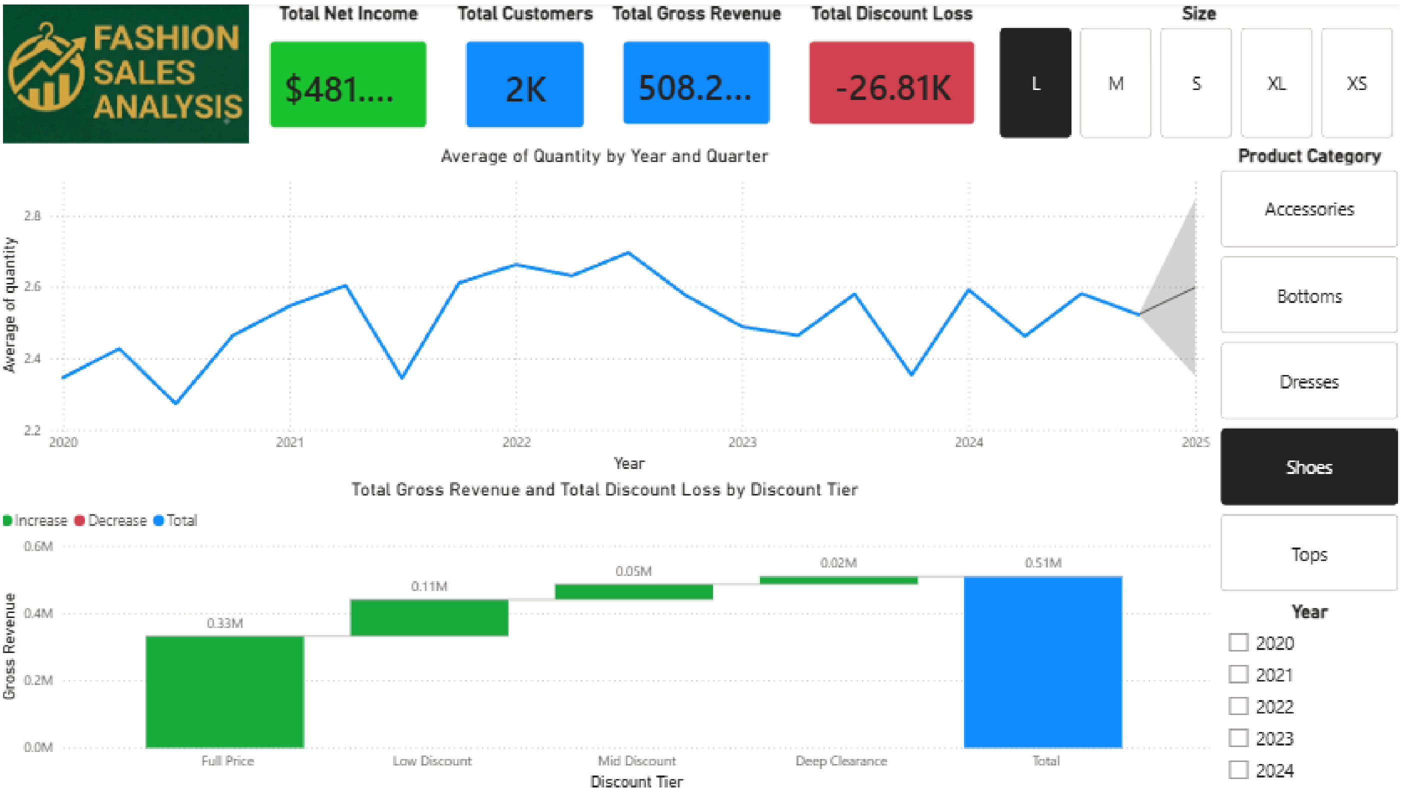


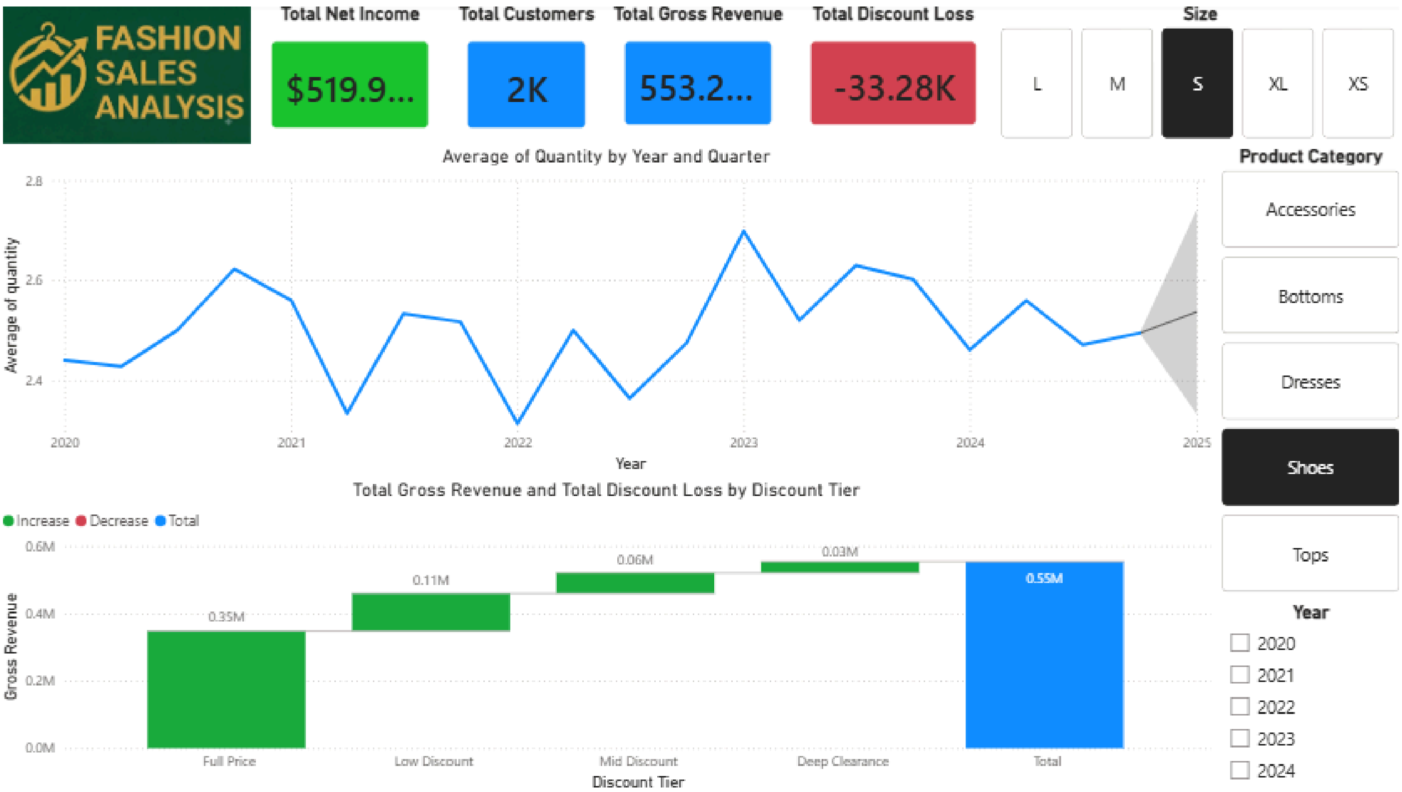
Dresses

- Stock up the following sizes: L

Shoes

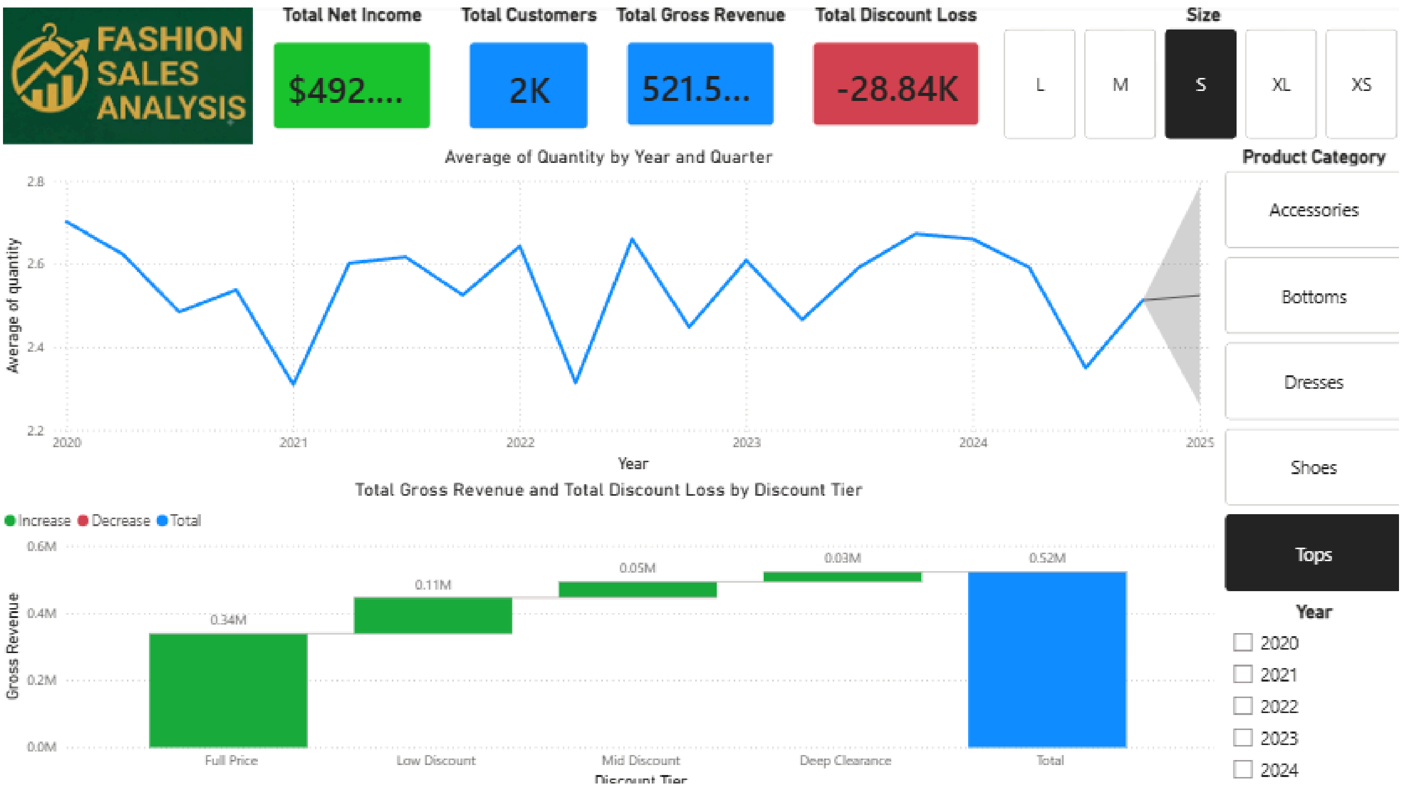
- Stock up the following sizes: L, S

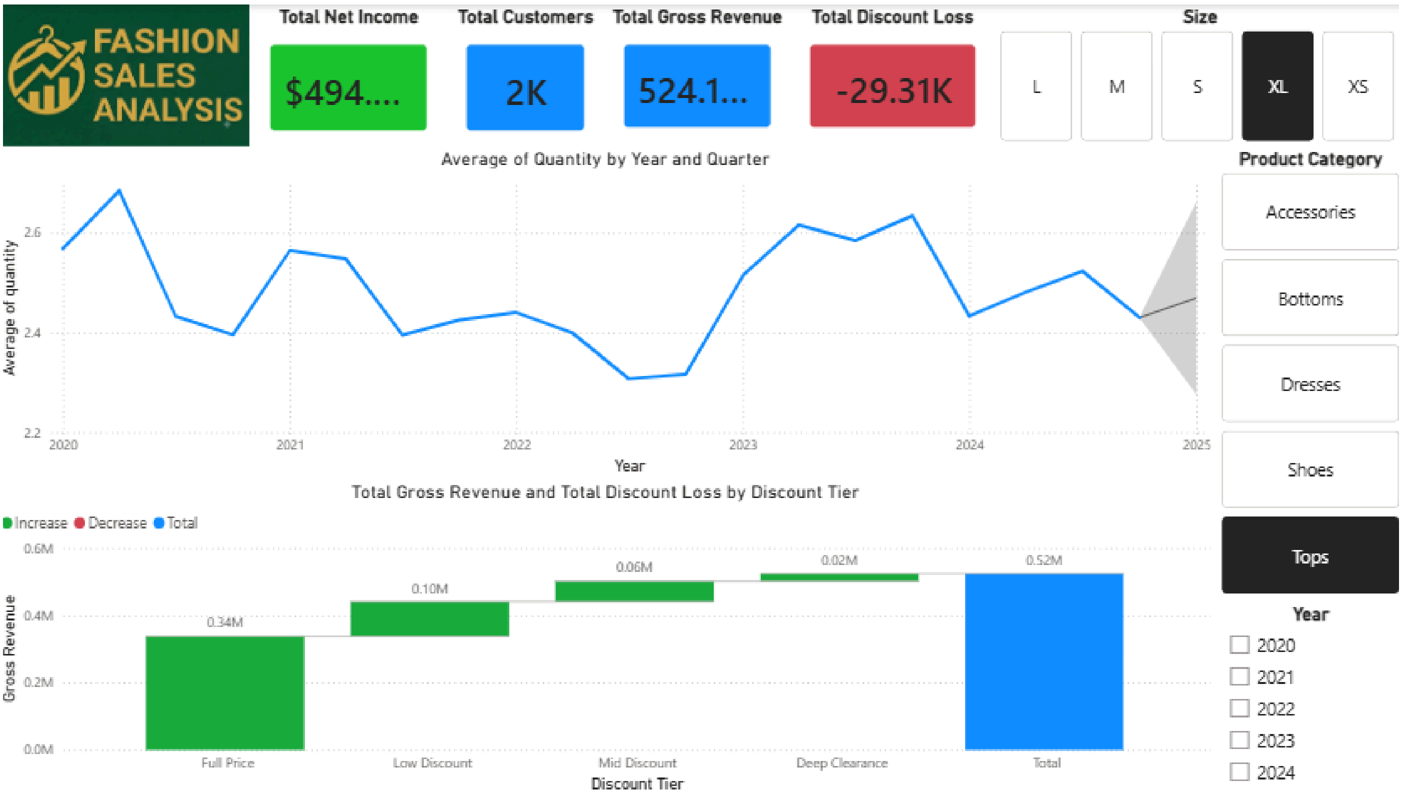


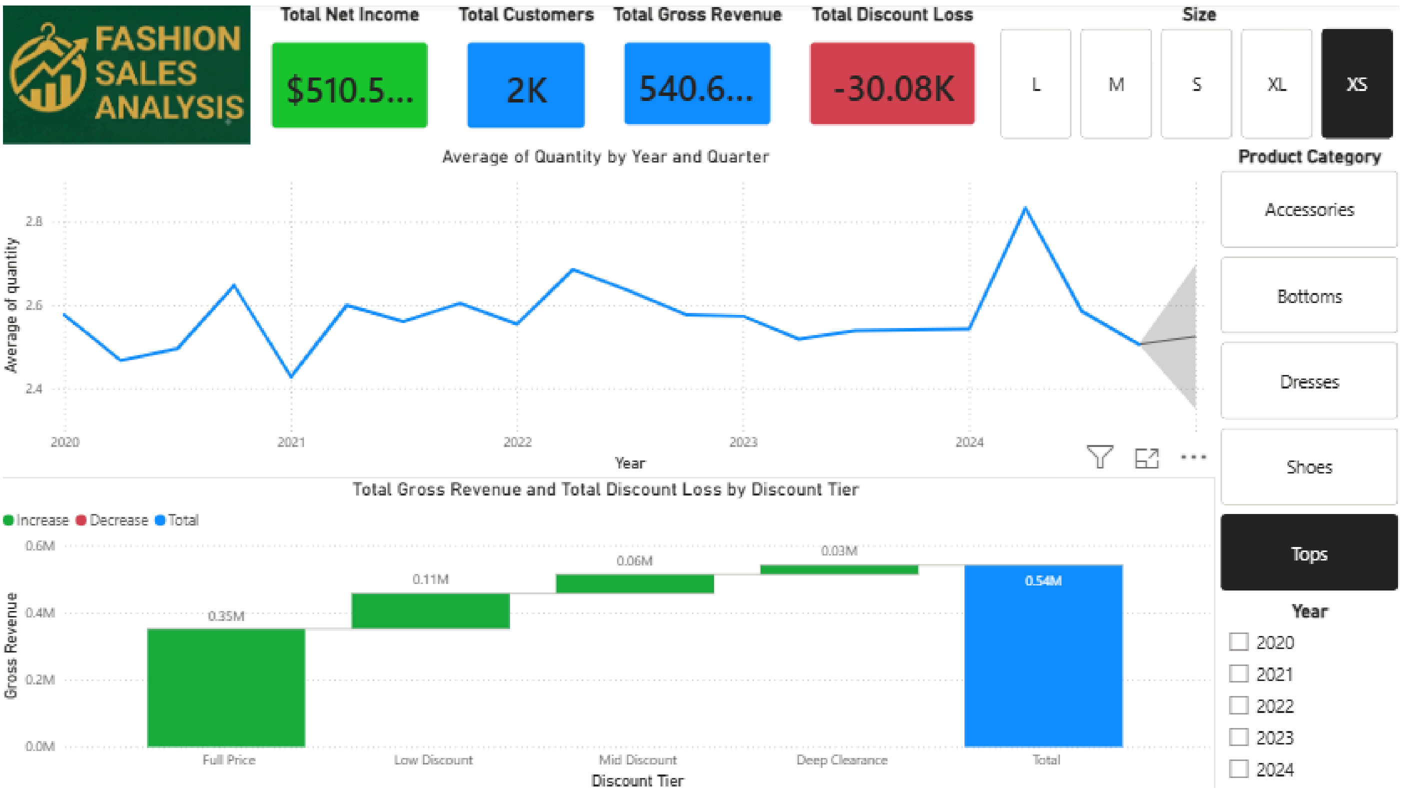


Tops

- Stock up the following sizes: M, S, XL, XS


FASHION SALES ANALYSIS
Total Net Income
$484....
Total Customers
2K
Total Gross Revenue
513.0...
Total Discount Loss
-28.70K
Size
L
M
S
XL
XS
Average of Quantity by Year and Quarter
Average of quantity
2.8
2.6
2.4
2.2
2020
2021
2022
2023
2024
2025
Year
Product Category
Accessories
Bottoms
Dresses
Shoes
Tops
Total Gross Revenue and Total Discount Loss by Discount Tier
Increase
Decrease
Total
Gross Revenue
0.6M
0.4M
0.2M
0.0M
0.34M
0.09M
0.05M
0.03M
0.51M
Full Price
Low Discount
Mid Discount
Deep Clearance
Total
Discount Tier
Year
2020
2021
2022
2023
2024

FASHION SALES ANALYSIS
Total Net Income
$492....
Total Customers
2K
Total Gross Revenue
521.5...
Total Discount Loss
-28.84K
Size
L
M
S
XL
XS
Average of Quantity by Year and Quarter
Average of quantity
2.8
2.6
2.4
2.2
2020
2021
2022
2023
2024
2025
Year
Product Category
Accessories
Bottoms
Dresses
Shoes
Tops
Total Gross Revenue and Total Discount Loss by Discount Tier
Increase
Decrease
Total
Gross Revenue
0.6M
0.4M
0.2M
0.0M
0.34M
0.11M
0.05M
0.03M
0.52M
Full Price
Low Discount
Mid Discount
Deep Clearance
Total
Year
2020
2021
2022
2023
2024





b. How should we structure our automated markdown timeline to liquidate slow-moving stock before its value plummets?

Possible Structure:

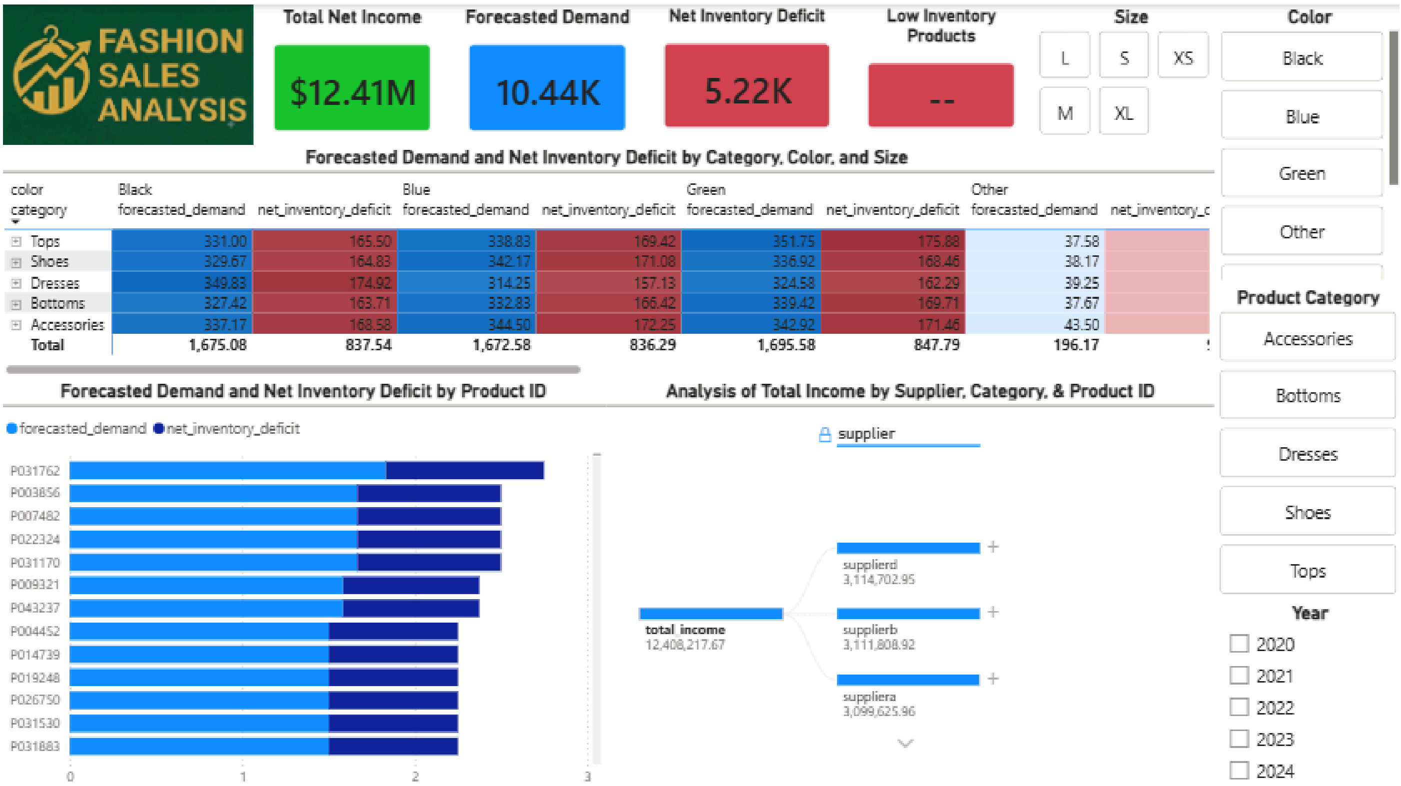
- **Days 1–30 (Full Price Tier):** Retain a 0% discount on new releases or peak seasonal items while quarterly sales velocity aligns with your target baseline trajectory.
- **Days 31–60 (Low Discount Tier):** Trigger an automated **10% discount** as soon as sales velocity drops below target to stimulate incremental revenue while minimizing discount loss.
- **Days 61–90 (Mid Discount Tier):** Escalate to a **20% discount** for items missing sell-through targets, targeting specific low-performing sizes (e.g., XS, XL) or categories via page slicers.

- **Days 90+ (Deep Clearance Tier):** Apply **Deep Clearance** pricing to liquidate remaining end-of-season stock near cost, recovering working capital and clearing floor space before value plummets.

3. Supply Chain & Supplier Performance
    a. Which specific variants are predicted to face stockouts or critical inventory deficits next month?

General Overview:

- There are NO Low Inventory Products so far. This is measured using the net_inventory_measure. This measure calculates the target_baseline_stock (calculated by finding the monthly_demand_rate and multiplying by 1.5) and subtracting the monthly_demand_rate.
- It can also be inferred that the forecasted net inventory deficit does not surpass the forecasted demand as well.



    b. How should we reallocate our upcoming procurement orders among top-performing vendors to prevent this?

Supplier Analysis:

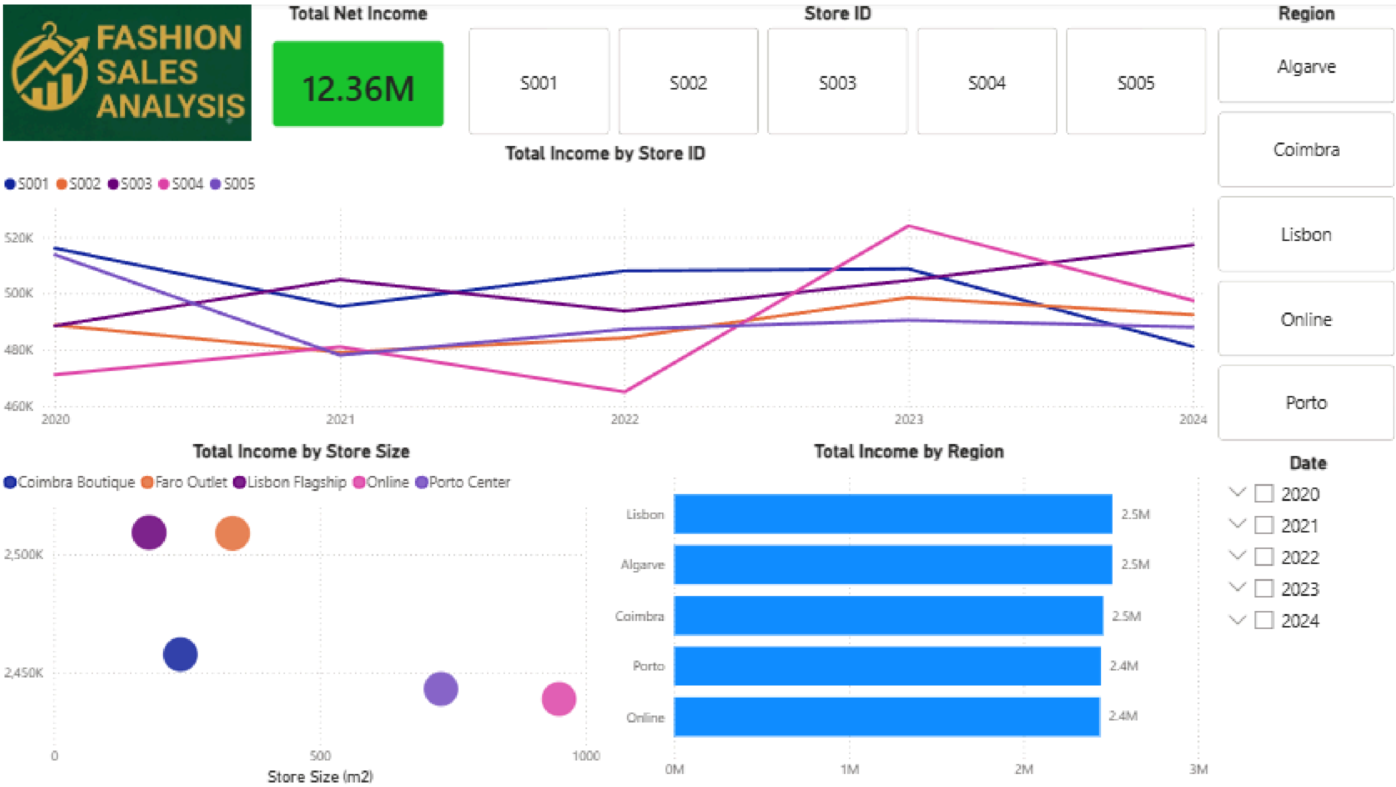
- All of the three suppliers (as seen above) are top-performing vendors. There is only a little margin between them in terms of total income.
- Align Category Strengths: Use your Decomposition Tree (supplier → category) to shift future orders for specific product lines to the vendor generating the highest income in that category.
- Leverage Vendor Metrics: Shift order volume toward vendors with higher delivery accuracy and shorter lead times to maintain safety stock levels.
- Apply 60/30/10 Split: Allocate 60% of primary orders to your top vendor, 30% to a secondary supplier as a fail-safe, and hold 10% for rapid re-orders.
- Automate Deficit Triggers: Set up dynamic procurement alerts to reallocate order volume to the fastest supplier as soon as [net_inventory_deficit] nears zero for any variant.

4. Omnichannel & Store Performance Benchmarking
    a. What is the total income per location?

General Overview:

- There is only a little difference in total net income per region. However, the income trend per store ID differs.

- Stores S001, S002, and S004 are experiencing a downward trend in total income starting in 2024. Store S004 has been on an upward trend since 2022. Store S005 seems to have been stagnant for the past 3 years.
- It can also be observed that the physical stores have higher total income compared to online by their store size. Physical stores are valuable in generating income, and it also means that the physical products are effective at attracting customers.
- Stores such as Lisbon Flagship and Faro Outlet have a smaller store size but have higher total income compared to the online store and other physical stores. This warrants further investigation into how the smaller stores have higher total income than others that have a bigger physical store.



b. Which physical stores should we proactively plan to transition into localized omnichannel fulfillment centers?

Suggested Approaches:

- Prioritize High-Footprint / Moderate-Revenue Stores: Look at your Total Income by Store Size scatter plot. Stores with large physical spaces (store_size_m2) but lower or declining walk-in revenue (e.g., S002 experiencing downward trends) are prime candidates to convert unused retail space into order-picking and staging areas.
- Leverage High-Performing Regional Hubs: Evaluate the Total Income by Region chart. Regions like Lisbon and Algarve generate high total income; positioning fulfillment centers in these top regions ensures fast local delivery and click-and-collect services for the largest customer base.
- Protect High-Yield Small Stores: Smaller locations with high sales density per square meter (e.g., Lisbon Flagship or Faro Outlet) should remain strictly focused on retail sales rather than taking up valuable footprint for inventory storage.
- Implement a Dual-Role (Hybrid) Pilot: Transition candidate stores incrementally—allocating backroom space for Ship-from-Store (SFS) and Buy-Online-Pick-Up-In-Store (BOPIS) services first before full conversion.